Laboratorio de Microeconomía I

Centro de Investigación y Docencia Económicas
Maestría en Economía - 2025
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Tarea 11

Exercises

Ex. 3.10 A *Leontief* production function has the form

$$y = \min\{\alpha x_1,\, \beta x_2\}$$

for $\alpha > 0$ and $\beta > 0$. Carefully sketch the isoquant map for this technology and verify that the elasticity of substitution $\sigma = 0$, where defined.

Ex. 3.11 Calculate σ for the Cobb–Douglas production function $y = Ax_1^{\alpha} x_2^{\beta}$, where $A > 0$, $\alpha > 0$, and $\beta > 0$.

Ex. 3.14 Calculate the elasticity of substitution for the production function in Example 3.2.

Ex. 3.21 What restrictions must there be on the parameters of the Cobb–Douglas form in Example 3.4 in order that it be a legitimate cost function?

Ex. 3.24 *True or false?* 'If $\lambda(\mathbf{w}, y)$ is the Lagrangian multiplier associated with the firm's cost-minimisation problem, then $mc(\mathbf{w}, y) = \lambda(\mathbf{w}, y)$.'

Ex. 3.26 Calculate the cost function and conditional input demands for the Leontief production function in Exercise 3.8.

Ex. 3.36 Derive the cost function for the two-input, constant-returns, Cobb–Douglas technology. Fix one input and derive the short-run cost function. Show that long-run average and long-run marginal cost are constant and equal. Show that for every level of the fixed input, short-run average cost and long-run average cost are equal at the minimum level of short-run average cost. Illustrate your results in the cost-output plane.

Ex. 5.C.10A Derive the cost function $c(w, q)$ and conditional factor demand functions (or correspondences) $z(w, q)$ for each of the following single-output constant return technologies with production functions given by

(a) $f(z) = z_1 + z_2$ (perfect substitutable inputs)

(b) $f(z) = \min\{z_1, z_2\}$ (Leontief technology)

(c) $f(z) = (z_1^\rho + z_2^\rho)^{1/\rho}, \quad \rho \leq 1$ (constant elasticity of substitution technology)